

Gaussian-Filtered Euclidean Propagators and UV Behaviour

Exact Domination under a Declared External Filter

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Abstract

The preceding delta-projection papers established a structural dictionary: many Dirac delta functions arise as sharp limits of finite kernels. This paper studies the Euclidean propagator obtained after declaring the external Gaussian filter

$$\Delta_\tau(k) = \frac{e^{-\tau|k|^2}}{|k|^2 + m^2}.$$

More generally, on one declared operator domain, let

$$B(A) = P\chi(A)e^{-\tau A/2}\chi(A)P.$$

This spectral window defines the compressed covariance

$$\Delta_B = B(A)L_m^{-1}B(A)^*.$$

We prove distributional recovery as $\tau \downarrow 0$, finite coincidence values, and Gaussian ultraviolet domination. Every scalar Euclidean graph whose internal lines are explicitly assigned this filter is absolutely ultraviolet finite under the stated quadratic-control hypothesis. This is a conditional model theorem, not a source theorem: an internal MTT gap does not select external momentum damping without an internal-to-external intertwiner. Reflection positivity, gauge invariance, causal support, and Lorentzian unitarity remain independent requirements.

Version 1 Revision Note

Supersedes The unversioned April 2026 manuscript.

Reason The earlier version promoted a familiar external heat filter from specified model data to a physically selected MTT propagator.

Resolution This version retains the exact Euclidean convergence and graph-domination theorems but treats the filter as a declared model input. It explicitly separates the missing source intertwiner, reflection positivity, gauge/BRST consistency, and Lorentzian unitarity.

Retained result All conditional Euclidean estimates remain available unchanged.

Remaining boundary A selected source theorem must emit the same external filter.

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1 Purpose and claim discipline

The earlier papers in this sequence established forward singular-limit statements of the form

$$K_\epsilon \longrightarrow \delta.$$

That is mathematically useful but not sufficient for prediction. The predictive direction is

$$\text{MTT fixed-point data} \implies K_{\text{MTT}} \implies \text{finite correction}.$$

Functional calculus selects a unique kernel once an operator, projector, domain, and scale have been declared. It does not identify that kernel with an external spacetime propagator. The present paper studies the consequences of making that additional model identification.

The paper proves a controlled statement in a flat Euclidean chart. It does not claim to complete an interacting physical QFT. It proves precise convergence, coincidence-finiteness, and perturbative domination properties for the displayed filter.

Status	Claim
Proved here	Gaussian-proper-time filtered propagators are finite at coincidence and recover the local propagator as $\tau \downarrow 0$.
Proved here	Euclidean scalar Feynman integrals with filtered internal lines are absolutely UV finite.
Conditional interpretation	τ may be an MTT coherence scale if a selected source intertwiner emits the external filter.
Not proved here	The numerical value of τ in a real physical sector.
Not proved here	Full Lorentzian unitarity, gauge invariance, or Standard Model phenomenology.

2 Novelty and non-novelty

The formula

$$\Delta_\tau(k) = \frac{e^{-\tau|k|^2}}{|k|^2 + m^2}$$

is not, by itself, a new regularization formula. Exponential and heat-kernel regulators are standard tools in mathematical physics and quantum field theory. The novelty claimed here is therefore not the bare functional form.

The possible MTT novelty is therefore a still-open selection statement:

selected MTT geometry emits this same external filter before amplitudes are evaluated.

In an MTT fixed-point regime, the data (A, P, χ, τ) have independent meaning before any loop integral is evaluated:

- A is the linearized damping/geometric operator around the coherent fixed point;
- P is the coherent-sector projector;
- $\chi(A)$ is the admissible spectral acceptance window associated with the gap and truncation regime;
- τ is the proper-time/coherence support scale of the evolve-project cycle.

Those meanings make the tuple a well-defined model input. They do not yet prove that its operator acts on external momentum or that the resulting covariance is the physical one.

Equivalently, the paper distinguishes three statements:

Statement	Status
The factor $e^{-\tau k^2}$ is a familiar heat-kernel filter	not novel
Using such a factor as a regulator	not novel by itself
Deriving (A, P, χ, τ) from MTT fixed-point data and then obtaining the filter	the execution-level novelty target

This paper proves the consequences of the third statement *conditional on* the required source intertwiner. It does not prove the selection statement itself.

3 Declared Euclidean benchmark input

Let X denote a flat coherent chart, either $\mathbb{R}^d$ or a torus T^d . Let

$$A = -\Delta$$

be the nonnegative chart Laplacian. Let P be a declared projector and let $\chi(A)$ be a declared spectral window. The associated filter is

$$B_{\tau,\chi} := P\chi(A)e^{-\tau A/2}\chi(A)P, \quad \tau > 0.$$

The corresponding kernel is

$$K_{\tau,\chi}(x, y) = \langle x | B_{\tau,\chi} B_{\tau,\chi}^* | y \rangle.$$

In the simplest full flat chart, $P = I$ and $\chi = I$, so

$$B_\tau = e^{-\tau A/2}, \quad K_\tau = e^{-\tau A}.$$

In momentum variables,

$$\widehat{K_\tau}(k) = e^{-\tau|k|^2}.$$

Thus K_τ is the heat-kernel replacement of the Dirac delta:

$$K_\tau(x - y) = (4\pi\tau)^{-d/2} \exp\left(-\frac{|x - y|^2}{4\tau}\right), \quad K_\tau \rightarrow \delta \quad (\tau \downarrow 0).$$

Remark 3.1 (Sector identity, not universal identity). If $P \neq I$, then the kernel is an identity only on the retained coherent sector. The limit $\tau \downarrow 0$ gives the sector kernel $P\chi(A)^2P$, not the full-space Dirac delta unless the sector/window tends to the full identity. This distinction is essential.

4 A worked declared circle example: S_R^1

To make the functional-calculus mechanism explicit, consider the chart $X = S_R^1$, the circle of radius R , with coordinate $\theta \in [0, 2\pi)$. Let

$$A = -R^{-2}\partial_\theta^2.$$

The eigenfunctions and eigenvalues are

$$\phi_n(\theta) = \frac{1}{\sqrt{2\pi R}} e^{in\theta}, \quad \lambda_n = \frac{n^2}{R^2}, \quad n \in \mathbb{Z}.$$

Once a proper-time scale $\tau_0 > 0$ is declared, functional calculus fixes

$$K_{\tau_0}^{S_R^1}(\theta, \theta') = \frac{1}{2\pi R} \sum_{n \in \mathbb{Z}} e^{-\tau_0 n^2 / R^2} e^{in(\theta - \theta')}.$$

The corresponding scalar propagator with mass $m > 0$ is

$$\Delta_{\tau_0}^{S_R^1}(\theta, \theta') = \frac{1}{2\pi R} \sum_{n \in \mathbb{Z}} \frac{e^{-\tau_0 n^2 / R^2}}{n^2 / R^2 + m^2} e^{in(\theta - \theta')}.$$

Relative to the declared model, the finite correction is fully determined by

$$(R, A, \tau_0).$$

There is no further freedom after that triple is fixed. The example does not select the triple or identify the circle operator with external spacetime momentum.

The effective spectral width is read off from

$$e^{-\tau_0 n^2 / R^2} \sim e^{-1} \iff |n| \sim \frac{R}{\sqrt{\tau_0}},$$

or, in physical momentum units,

$$p_{\text{eff}} \sim \tau_0^{-1/2}.$$

This is the simplest explicit model of the conditional claim:

declared operator data determine a unique finite kernel.

5 Gaussian-filtered scalar propagator

Let

$$L_m := A + m^2 = -\Delta + m^2, \quad m > 0.$$

The ordinary Euclidean propagator is

$$\Delta_0 := L_m^{-1}.$$

The filtered covariance is

$$\Delta_{\tau, \chi} := B_{\tau, \chi} L_m^{-1} B_{\tau, \chi}^*.$$

In the full flat chart $P = \chi = I$, this becomes

$$\Delta_\tau = e^{-\tau A/2} (A + m^2)^{-1} e^{-\tau A/2} = e^{-\tau A} (A + m^2)^{-1}.$$

Hence in momentum space,

$$\widehat{\Delta}_\tau(k) = \frac{e^{-\tau|k|^2}}{|k|^2 + m^2}. \quad (5.1)$$

Definition 5.1 (Effective filter scale). The proper-time parameter $\tau > 0$ defines the effective coherence scale

$$\Lambda_{\text{eff}} := \tau^{-1/2}.$$

Modes with $|k| \gg \Lambda_{\text{eff}}$ are exponentially suppressed.

6 Recovery of the ordinary propagator

Theorem 6.1 (Distributional recovery). *Let $m > 0$, $d < \infty$, and define Δ_τ by [equation \(5.1\)](#). Then*

$$\Delta_\tau \longrightarrow \Delta_0$$

in the sense of tempered distributions as $\tau \downarrow 0$. Equivalently, for all Schwartz functions $f, g \in \mathcal{S}(\mathbb{R}^d)$,

$$\lim_{\tau \downarrow 0} \int_{\mathbb{R}^d} \frac{e^{-\tau|k|^2}}{|k|^2 + m^2} \widehat{f}(k)^* \widehat{g}(k) \frac{d^d k}{(2\pi)^d} = \int_{\mathbb{R}^d} \frac{\widehat{f}(k)^* \widehat{g}(k)}{|k|^2 + m^2} \frac{d^d k}{(2\pi)^d}.$$

Proof. For each k , $e^{-\tau|k|^2} \rightarrow 1$. Moreover,

$$0 \leq \frac{e^{-\tau|k|^2}}{|k|^2 + m^2} \leq \frac{1}{|k|^2 + m^2}.$$

Since f, g are Schwartz, the function

$$\frac{|\widehat{f}(k)| |\widehat{g}(k)|}{|k|^2 + m^2}$$

is integrable. Dominated convergence gives the result. $\square$

7 Finite coincident value

The ordinary coincident propagator

$$\Delta_0(x, x) = \int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} \frac{1}{|k|^2 + m^2}$$

diverges for $d \geq 2$. The filtered propagator has

$$\Delta_\tau(x, x) = I_d(m, \tau) := \int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} \frac{e^{-\tau|k|^2}}{|k|^2 + m^2}.$$

Theorem 7.1 (Finite diagonal). *For every $d < \infty$, $m > 0$, and $\tau > 0$,*

$$I_d(m, \tau) < \infty.$$

Moreover,

$$I_d(m, \tau) = \frac{1}{(4\pi)^{d/2}} \int_\tau^\infty u^{-d/2} e^{-m^2 u} du.$$

Proof. Use

$$\frac{1}{|k|^2 + m^2} = \int_0^\infty e^{-s(|k|^2 + m^2)} ds.$$

Then

$$I_d(m, \tau) = \int_0^\infty e^{-m^2 s} \left[\int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} e^{-(s+\tau)|k|^2} \right] ds.$$

The Gaussian integral gives

$$\int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} e^{-(s+\tau)|k|^2} = (4\pi(s+\tau))^{-d/2}.$$

With $u = s + \tau$,

$$I_d(m, \tau) = \frac{1}{(4\pi)^{d/2}} \int_\tau^\infty u^{-d/2} e^{-m^2(u-\tau)} du.$$

If the filter is placed as $e^{-\tau A/2}$ on each side, the propagator carries $e^{-\tau|k|^2}$, giving the harmless factor $e^{m^2\tau}$ depending on the chosen proper-time normalization. Equivalently one may define the shifted proper-time representation

$$\frac{e^{-\tau|k|^2}}{|k|^2 + m^2} = e^{m^2\tau} \int_\tau^\infty e^{-u(|k|^2 + m^2)} du.$$

Thus

$$I_d(m, \tau) = \frac{e^{m^2 \tau}}{(4\pi)^{d/2}} \int_{\tau}^{\infty} u^{-d/2} e^{-m^2 u} du.$$

Both forms are finite for $\tau > 0$. The displayed normalization depends on whether the mass term is included in the heat filter. In either convention the lower proper-time gap $\tau > 0$ removes the $u = 0$ singularity. $\square$

Remark 7.2 (Normalization convention). Two closely related filters occur in the literature:

$$e^{-\tau A}(A + m^2)^{-1}, \quad e^{-\tau(A+m^2)}(A + m^2)^{-1}.$$

They differ by the finite factor $e^{-\tau m^2}$. All ultraviolet estimates are unchanged. For a proper-time representation the second convention is often cleaner. Neither convention by itself proves Osterwalder–Schrader reflection positivity.

8 Leading low-energy correction

For $\tau|k|^2 \ll 1$,

$$e^{-\tau|k|^2} = 1 - \tau|k|^2 + \frac{\tau^2|k|^4}{2} + O(\tau^3|k|^6).$$

Therefore

$$\widehat{\Delta}_{\tau}(k) = \frac{1}{|k|^2 + m^2} - \tau \frac{|k|^2}{|k|^2 + m^2} + O\left(\tau^2 \frac{|k|^4}{|k|^2 + m^2}\right).$$

Proposition 8.1 (Low-momentum correction bound). *For $|k|^2 \leq R^2$,*

$$\left| \frac{e^{-\tau|k|^2} - 1}{|k|^2 + m^2} \right| \leq \frac{\tau|k|^2}{|k|^2 + m^2} \leq \tau.$$

Thus the pointwise low-energy correction is $O(\tau|k|^2)$ relative to the local numerator and bounded by $O(\tau)$ in absolute propagator units on compact momentum sets.

Proof. Use $0 \leq 1 - e^{-a} \leq a$ for $a \geq 0$. $\square$

9 One-loop examples

9.1 Tadpole

The ordinary one-loop tadpole integral is

$$T_0 = \int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} \frac{1}{|k|^2 + m^2},$$

which is divergent for $d \geq 2$. The MTT-filtered tadpole is

$$T_{\tau} = \int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} \frac{e^{-\tau|k|^2}}{|k|^2 + m^2},$$

which is finite by the finite-diagonal theorem.

9.2 Bubble

For external momentum p ,

$$B_\tau(p) = \int_{\mathbb{R}^d} \frac{d^d k}{(2\pi)^d} \frac{e^{-\tau|k|^2} e^{-\tau|k+p|^2}}{(|k|^2 + m^2)(|k+p|^2 + m^2)}.$$

Proposition 9.1 (Bubble finiteness). *For every $p \in \mathbb{R}^d$, $m > 0$, and $\tau > 0$,*

$$B_\tau(p) < \infty.$$

Proof. For large $|k|$, $|k+p|^2 \geq \frac{1}{2}|k|^2 - |p|^2$. Hence

$$e^{-\tau|k|^2} e^{-\tau|k+p|^2} \leq C_p e^{-c_\tau|k|^2}$$

for some $C_p, c_\tau > 0$. The denominator is positive and polynomial, so the integrand is dominated by an integrable Gaussian. $\square$

10 All-graph Euclidean UV domination

We now state a controlled perturbative result for scalar Euclidean diagrams. The point is not to construct a full QFT, but to show that the declared filtering gives absolute ultraviolet convergence once every internal line carries the Gaussian factor.

Consider a connected graph Γ with L loop momenta $\ell = (\ell_1, \dots, \ell_L) \in (\mathbb{R}^d)^L$, internal lines $e \in E(\Gamma)$, and fixed external momenta. Each internal line momentum has the affine form

$$q_e(\ell) = M_e \ell + r_e,$$

where M_e is a linear map and r_e depends on external momenta. The filtered scalar integrand is bounded by

$$\prod_{e \in E(\Gamma)} \frac{e^{-\tau|q_e(\ell)|^2}}{|q_e(\ell)|^2 + m^2}$$

times a polynomial numerator $P(\ell)$, if derivative interactions are present.

Lemma 10.1 (Loop-momentum quadratic control). *For a connected graph and a choice of independent loop momenta, there exist constants $c_\Gamma > 0$ and $C_\Gamma \geq 0$, depending on the graph and external momenta, such that*

$$\sum_{e \in E(\Gamma)} |q_e(\ell)|^2 \geq c_\Gamma |\ell|^2 - C_\Gamma.$$

Proof. If the inequality failed, there would be a sequence $|\ell_n| \rightarrow \infty$ with all line momenta $q_e(\ell_n)$ sublinear in $|\ell_n|$. After normalizing, a nonzero loop direction would lie in the kernel of every line-momentum map M_e . But independent loop momenta are chosen precisely so that each nonzero loop flow crosses at least one internal line. Hence the common kernel is trivial. Finite-dimensional norm equivalence then gives the stated coercivity, with the affine external shifts absorbed into C_Γ . $\square$

Theorem 10.2 (Euclidean all-graph UV finiteness under declared line filtering). *Fix $m > 0$, $\tau > 0$, and a connected scalar Euclidean Feynman graph Γ . Suppose every internal line carries the declared Gaussian factor $e^{-\tau|q_e|^2}$. Then the loop integral*

$$\int_{(\mathbb{R}^d)^L} |P(\ell)| \prod_{e \in E(\Gamma)} \frac{e^{-\tau|q_e(\ell)|^2}}{|q_e(\ell)|^2 + m^2} d^d \ell$$

is finite for every polynomial numerator P .

Proof. By the quadratic-control lemma,

$$\prod_e e^{-\tau|q_e(\ell)|^2} = e^{-\tau \sum_e |q_e(\ell)|^2} \leq e^{\tau C_\Gamma} e^{-\tau c_\Gamma |\ell|^2}.$$

The denominator is bounded above by dropping it, since $(|q_e|^2 + m^2)^{-1} \leq m^{-2}$. Thus the absolute integrand is bounded by

$$C|P(\ell)|e^{-\tau c_\Gamma |\ell|^2},$$

which is integrable on finite-dimensional loop-momentum space. $\square$

Remark 10.3 (Physical compatibility boundary). This theorem is Euclidean and scalar. Gauge theories require compatible filtering of fields, ghosts, vertices, and Ward/BRST identities. Lorentzian unitarity requires a separate continuation or positive spectral representation, and Euclidean use requires a reflection-positivity certificate. The theorem proves ultraviolet domination only.

11 Proper-time representation and positivity

For the mass-including convention,

$$\Delta_\tau^{(m)}(k) = \frac{e^{-\tau(|k|^2 + m^2)}}{|k|^2 + m^2},$$

one has the exact proper-time representation

$$\Delta_\tau^{(m)}(k) = \int_\tau^\infty e^{-s(|k|^2 + m^2)} ds.$$

This is a positive proper-time integral with a lower support gap $s \geq \tau$. The ordinary propagator is recovered by lowering the gap to zero:

$$\frac{1}{|k|^2 + m^2} = \int_0^\infty e^{-s(|k|^2 + m^2)} ds.$$

This representation gives the Euclidean model reading:

UV softening = absence of arbitrarily short proper-time support.

It is compatible with interpreting τ as finite coherent support, but does not derive that interpretation.

12 Regulator and the MTT source obligation

Mathematically, the factor $e^{-\tau|k|^2}$ is a standard UV filter. The stronger MTT claim would require:

a selected operator-domain intertwiner derives the external filter from MTT geometry.

Until that theorem is supplied, this paper is a regulator calculation with a precisely declared filter. Functional calculus makes the output unique relative to the input tuple, but it does not select that tuple or identify its domain with external momentum space.

An arbitrary smearing function may break locality, positivity, gauge symmetry, or unitarity. A physically admissible MTT kernel must be selected by the coherent projector, spectral window, proper-time support, and compatibility conditions of the relevant sector.

Object	Role
A	declared nonnegative Euclidean operator
P	declared retained-sector projector
$\chi(A)$	declared spectral window
τ	declared proper-time/filter scale
$B_{\tau,\chi}$	declared functional-calculus filter
$\Delta_{\tau,\chi}$	Gaussian-filtered covariance/propagator
$\Lambda_{\text{eff}} \sim \tau^{-1/2}$	effective UV/coherence scale

13 Numerical anchoring from existing data

The model does not yet derive τ numerically from the full MTT carrier. Nevertheless, once the propagator correction is fixed, existing data translate directly into bounds or target scales.

For low momenta,

$$e^{-\tau k^2} = 1 - \tau k^2 + O(\tau^2 k^4).$$

Thus an experiment that sees no deviation from the local propagator up to characteristic momentum scale E , with relative tolerance ε_{exp} , gives the parametric bound

$$\tau E^2 \lesssim \varepsilon_{\text{exp}}, \quad \Lambda_{\text{eff}} = \tau^{-1/2} \gtrsim \frac{E}{\sqrt{\varepsilon_{\text{exp}}}}.$$

This is deliberately written as a bound rather than a prediction, because the sector value of τ is not fixed here.

As an illustrative collider-scale anchoring, LHC Run 3 operates at a proton–proton collision energy of 13.6 TeV [5]. Requiring the coherence scale to lie above this scale gives the conservative estimate

$$\Lambda_{\text{eff}} \gtrsim 13.6 \text{ TeV}, \quad \sqrt{\tau} \lesssim 1.45 \times 10^{-20} \text{ m}.$$

A stronger illustrative benchmark comes from contact-interaction/compositeness searches, where global summaries commonly quote lower scales in the multi-TeV to tens-of-TeV range [6]. Taking

$$\Lambda_{\text{eff}} \gtrsim 20 \text{ TeV} \quad \text{or} \quad 50 \text{ TeV}$$

corresponds to

$$\sqrt{\tau} \lesssim 9.87 \times 10^{-21} \text{ m}, \quad \sqrt{\tau} \lesssim 3.95 \times 10^{-21} \text{ m},$$

respectively. These numbers should not be read as established MTT predictions. They show how existing high-energy data would bound the coherence length once the filtered-propagator correction is identified with a physical sector.

For comparison, the 2022 CODATA Planck length is

$$\ell_{\text{P}} = 1.616255(18) \times 10^{-35} \text{ m}$$

[7]. A Planck-scale coherence length would correspond to

$$\Lambda_{\text{eff}} \sim 1.22 \times 10^{19} \text{ GeV},$$

far beyond present collider reach. The present framework allows either possibility: τ may be experimentally bounded, or it may be fixed by deeper MTT carrier data at a much smaller length scale.

Benchmark	Λ_{eff}	$\sqrt{\tau} \sim \Lambda_{\text{eff}}^{-1}$
LHC Run 3 collision scale	13.6 TeV	1.45×10^{-20} m
Illustrative contact scale	20 TeV	9.87×10^{-21} m
Illustrative contact scale	50 TeV	3.95×10^{-21} m
Planck length benchmark	1.22×10^{19} GeV	1.62×10^{-35} m

The important point is conceptual: the filtered propagator converts any empirical lower bound on unresolved short-distance structure into a bound on the MTT coherence scale,

$$\sqrt{\tau} \lesssim \Lambda_{\text{eff}}^{-1}.$$

14 What becomes predictive

The paper proves that, once A, P, χ, τ and the external-domain identification are declared, the propagator correction is fixed. The remaining execution tasks are concrete:

$$(A, P, \chi, \tau) \implies \Delta_{\tau, \chi} \implies \text{observable correction.}$$

Examples include:

1. finite coincident correlators;
2. modified short-distance potentials;
3. exponentially softened loop amplitudes;
4. low-energy corrections of order $\tau|k|^2$;
5. effective scale constraints $\Lambda_{\text{eff}} \sim \tau^{-1/2}$.

The crucial unresolved step is to remove the word “declared”: the corrected propagator must be emitted by a selected source theorem rather than chosen for its ultraviolet behavior.

15 Conclusion

The delta-projection program requires a transition from interpretation to execution. The canonical coherent-kernel paper supplies a conditional functional-calculus construction. This paper evaluates its external Gaussian benchmark, without claiming the missing physical source theorem.

In a flat Euclidean chart, the declared filter gives

$$\widehat{\Delta}_{\tau}(k) = \frac{e^{-\tau|k|^2}}{|k|^2 + m^2}.$$

This propagator recovers the ordinary local propagator as $\tau \downarrow 0$, has finite coincident value in every finite dimension when $\tau > 0$, and gives all-graph Euclidean ultraviolet domination for scalar perturbation theory when every internal line carries the declared filter.

The result neither derives τ nor proves that an MTT fixed-point sector supplies this external momentum filter. It shows what follows if it does. Existing high-energy data can be translated into bounds on $\sqrt{\tau}$ only under that same physical identification.

The Euclidean correction is exact once the external filter is declared; MTT selection remains open.

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